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Mini Project Proposal: FIFA Football Match Predictor

Submission Window: 3 Aug – 7 Aug

1. Refined Problem Definition (Abstract)

Predicting football match outcomes is valuable for analysts, coaches, and fans, but most existing models rely only on **team-level statistics** (rankings, past results) and ignore the specific players on the field. This is a major gap for **FIFA World Cup prediction** specifically, since national squads are rebuilt every tournament from players across different clubs — making team reputation alone an unreliable predictor.

This project aims to build a **machine learning-based match outcome predictor** that combines: - **Player-level data** — goals, assists, passing accuracy, tackles, physical/mental attributes - **Team-level data** — historical results, head-to-head record, rankings

The system will use **feature scaling + dimensionality reduction (PCA)**, followed by an **ensemble of classifiers** (Logistic Regression, Random Forest, XGBoost, AdaBoost, KNN) combined through **majority voting**, to predict the winner of a football match with higher accuracy than a simple historical-record baseline.

Objective: Build a generalizable, player-aware ML model for World Cup match prediction and benchmark it against a baseline (historical win-ratio) model.

Scope: FIFA World Cup matches (2015–2023 player/team data), binary win/loss classification.

2. Hardware Requirements

Component	Requirement	Notes
Processor	Intel i5 / AMD Ryzen 5 or higher	Sufficient for tabular ML models
RAM	8 GB minimum (16 GB recommended)	Handles PCA + ensemble training comfortably
Storage	5–10 GB free space	For datasets, model files, notebooks
GPU	Not required	Optional only if extending to Neural Nets/GNNs later
Internet	Required	For dataset downloads and cloud notebooks

No specialized hardware is needed — a standard laptop is sufficient since the models are tabular/tree-based, not deep learning-heavy.

3. Software Requirements

Category	Tools
Language	Python 3.x
IDE / Environment	Jupyter Notebook or Google Colab
Data Handling	pandas, NumPy
ML Libraries	scikit-learn, XGBoost
Dimensionality Reduction	scikit-learn (PCA)
Visualization	Matplotlib, Seaborn
Version Control	Git + GitHub
Data Sources	Kaggle (FIFA player stats datasets), public World Cup match history datasets
Optional (future scope)	PyTorch / TensorFlow, PyTorch Geometric (for GNN extension)

All tools are free and open-source — no licensed software required.

4. Budget Submission

Item	Estimated Cost	Notes
Software/Tools	\$0	Python, scikit-learn, XGBoost — all free/open-source
Datasets	\$0	Public datasets (Kaggle, FIFA stats)
Cloud Compute	\$0 (Free tier)	Google Colab free tier sufficient
Optional Cloud GPU	~\$0–10	Only if experimenting with deep learning/GNN extension
Miscellaneous	\$0	No hardware purchase required
Total Estimated Budget	\$0 – \$10	Effectively free; minor optional cloud cost only

Phase	Task	Duration
7	Report Writing & Documentation	21 Sep – 25 Sep
8	Final Presentation / Demo	28 Sep – 2 Oct

Milestone summary: - [Done] Week 1: Proposal + Literature Survey (this submission) - [Pending] Weeks 2–4: Data pipeline + feature engineering - [Pending] Weeks 5–6: Model building + evaluation - [Pending] Weeks 7–8: Reporting + final demo